COIL DESIGN & ELECTROMAGNETIC SPECIFICATIONS

Sensor Front-End Engineering Addendum — Pulse Induction System

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1. Design Overview

The electromagnetic sensor front-end relies on a discrete, two-coil air-core setup optimized for a Pulse Induction (PI) switching matrix. Unlike single-coil systems which experience severe signal degradation due to back-EMF switching noise leaking into the input amplification stage, this design isolates the primary pulse generation path from the secondary high-gain receive path. Both loops are wrapped utilizing professional-grade insulated copper wire over heavy-duty solenoidal forms to guarantee structural symmetry and stable baseline coefficients.

2. Comprehensive Structural Specifications

The geometric profiles, physical wire metrics, and turns density profiles have been meticulously implemented to establish clear inductance separation. The design matrix is detailed in the technical summary below:

Engineering Parameter	Transmitter Coil (TX)	Receiver Coil (RX)
Conductor Material	Enameled Copper Wire	Enameled Copper Wire
Conductor Wire Diameter	0.5 mm (~24 AWG)	0.5 mm (~24 AWG)
Total Windings Count (N)	**120 turns**	**50 turns**
Winding Tube Outer Diameter (d)	4.0 cm (1.575 inches)	5.0 cm (1.969 inches)
Total Winding Width / Length (l)	9.0 cm (3.543 inches)	5.0 cm (1.969 inches)
Calculated Target Inductance (L)	**≈ 210.0 µH**	**≈ 84.9 µH**

3. Mathematical Modeling & Expected Inductance

To approximate the expected electrical response parameters prior to analytical testing, Wheeler's classic equation for single-layer/low-layer air-core solenoids is used:

$$L\ (\mu H) = \frac{d^2 \times N^2}{18d + 40l}$$

Where d equals the coil diameter in inches, l represents the exact winding width/length in inches, and N indicates the complete turn count.

Transmitter Coil Evaluation: Given values $d = 1.575$ inches, $l = 3.543$ inches, and $N = 120$:

$L_{TX} = (1.575^2 \times 120^2) / ((18 \times 1.575) + (40 \times 3.543)) \approx 210.0\ \mu H$

Receiver Coil Evaluation: Given values $d = 1.969$ inches, $l = 1.969$ inches, and $N = 50$:

$L_{RX} = (1.969^2 \times 50^2) / ((18 \times 1.969) + (40 \times 1.969)) \approx 84.9\ \mu H$

4. Photographic Documentation

The physical layout, construction precision, and structural alignment of the wound forms are captured in the laboratory imagery below:

4.1 Primary Transmitter Coil Assembly

The primary driver element features a long-form solenoidal pattern designed to compress the flux profile, forcing a directional magnetic transient downward toward the material testing jig.

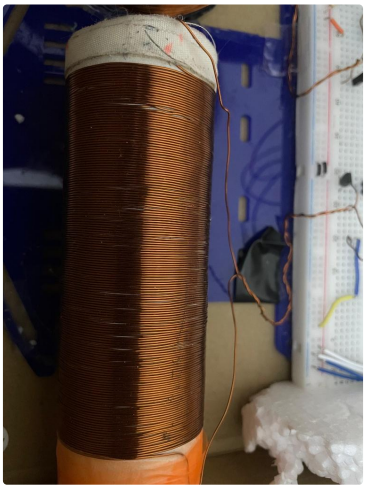


Figure 1: 120-turn primary Transmitter (TX) coil wound uniformly over a 4cm non-magnetic support form.

4.2 Secondary Receiver Coil Assembly

The receiver loop utilizes a wider, low-profile configuration to wrap cleanly around the core fields, allowing optimized coaxial nesting with high inductive cross-coupling parameters.



Figure 2: 50-turn secondary Receiver (RX) coil built over a 5cm outer diameter tube shell.

4.3 Coaxial Concentric Sensor Node

The nested configuration provides a compact sensor shell. This alignment minimizes baseline leakage spikes while focusing the eddy-current decay profiles directly into the high-speed LM358 conditioning network.



Figure 3: Dual-coil nested assembly showcasing concentric geometry for localized material interrogation.